Space Weather Events

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Aims:

- Identify space weather events using magnetograms and Kp data
- Estimate CME speed and arrival time from a series of coronagraph images
- Compare your estimations to the actual CME event and consider sources of error in your calculations

Materials:

- Paper copy of the document CME_Image_Set.docx
- Ruler
- Pencil
- Calculator

Space weather events are events driven by distant solar activity that impact human technology and the near-Earth space environment (BoM SWS). In this class, you will use real scientific data and plots to identify space weather events and then study one of these as a case study to calculate the speed and arrival time of the associated coronal mass ejection.

Activity 1: Identifying space weather events using magnetograms & Kp data

You have been provided with 3 sets of plots in the document **Space_Weather_Events_Plots.pdf**.

These correspond to 3 different 2-week time-periods:

A: 03-16 May 2024

B: 10-23 March 2015

C: 01-14 November 2019

The top plots for each time-period are **magnetograms**. Magnetograms are time-series plots that show variations in Earth's magnetic field in three orthogonal directions – x (north-south), y (east-west) and z (up-down) (BoM SWS). Measurements are commonly reported in nanotesla (nT) after Serbian-American inventor Nikola Tesla (Ungvarsky, 2023).

Geomagnetic storms are seen as **spikes** in magnetogram plots i.e. the lines suddenly fall/ rise to a minimum/ maximum value and then return to relatively flat (quiet) conditions.

The **x component** of magnetograms refers to geomagnetic disturbances in the magnetic north direction. This component is most relevant for identifying geomagnetic activity as geomagnetic storms cause temporary shifts in the position of magnetic north (NASA, 2015).

The bottom plots are time series plots of **Kp index** value. The Kp index is a measure of global geomagnetic disturbance on a scale from 0-9 taken at 3-hour intervals (SWPC NOAA). According to the NOAA space weather scale, a minimum Kp value of **5** is needed to classify an event as a geomagnetic storm.

For each event answer the following questions:

1. Was there a geomagnetic storm?
2. How do you know if there was a storm (or not)?
3. What date was peak geomagnetic disturbance?
4. Quantify peak geomagnetic disturbance (x component)/ nt
5. Max Kp index value reached
6. NOAA G-scale classification
7. What impacts do you think this event had?

These questions are listed in **Table 1** below.

You will need to refer to the NOAA space weather scales (available here or in the document **NOAAscales.pdf**) to answer the questions. This document contains information on the predicted impacts of space weather events at each level on the scale.

Table 1: Interpretations of the geomagnetic conditions for events A, B and C

	A 03-16 May 2024	**B** 10-23 March 2015	**C** 01-14 November 2019
Was there a geomagnetic storm?			
How do you know if there was a storm (or not)?			
What date was peak geomagnetic disturbance?			
Quantify peak geomagnetic disturbance (x component)/ nt HINT: calculate the max-min x value			
Max Kp index value reached			
NOAA G-scale classification			
What impacts do you think this event had?			

We are now going to consider the March 2015 event as a case study.

Activity 2: Calculating the speed & arrival time of the St Patrick's Day 2015 CME

Event B (March 2015) was a geomagnetic storm event named the **St. Patrick's Day solar storm**. We will be analysing the motion of the CME associated with the St Patrick's Day storm using a series of coronagraph images from NASA's Solar and Heliospheric Observatory (**SOHO**). The images are found in the file **CME_Image_Set.docx**.

Coronagraphs are specialised space-based instruments designed to block out the Sun's light so the corona (the Sun's outermost layer) can be viewed (Mann, 2019). Images from coronagraphs are used to monitor space weather events like coronal mass ejections (CMEs).

The white circle on each image is the outline of the Sun. The solid red disk is a suspended plate that blocks out the most intense sunlight, allowing SOHO to perceive the CMEs. A date and time stamp (in UT) can be found in the bottom left-hand corner.

The SOHO video recording of this CME can be found here.

1. On each image, mark a feature of the CME that you can trace between images. The shape of the CME changes over time like a cloud, so try to mark a feature at the front of the CME.

2. With a ruler, measure the distance (in mm) between the CME feature that you are tracking and the edge of the Sun for each image (see **Figure 1** as an example).

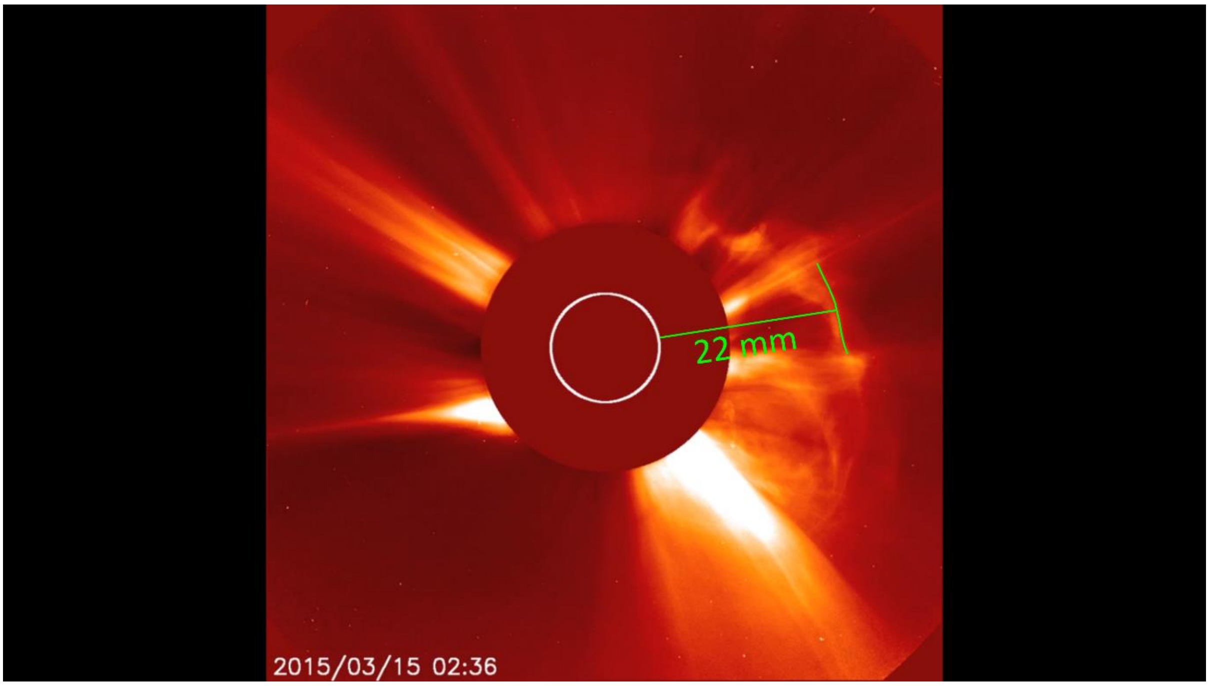

Figure 1: Example CME measurement (not to scale). Image courtesy of SOHO/LASCO consortium. SOHO is a project of international cooperation between ESA and NASA.

3. Add these positions to **Table 2** below and fill out the time columns.

Table 2: St Patrick's Day 2015 CME position, time and velocity calculated from the coronagraph images

Image Number	Position (mm)	Clock time	Time between images (hrs)	Total elapsed time (hrs)	Average velocity (mm/hr)
1					
2					
3					
4					
5					

4. Calculate the CME velocity for each image using v = s/t, where s is the position (mm) and t is the total elapsed time (hrs). Add these values to **Table 2**.

The velocity of the CME varies through time as the CME cloud expands as it moves towards Earth. We will take the velocity you calculated for **Image Number 5** as the velocity of the CME as it travels towards Earth. Make a note of this velocity (including units).

While using mm/hr is a convenient way to study general motion, it is more meaningful to convert these to km/hr in order to estimate how long it will take the CME to reach Earth.

5. Measure the diameter of the Sun in your images in mm. The actual diameter of the Sun is **1,391,400 km**. How many km is represented by one mm? This is your scale.

6. Convert your velocity for Image Number 5 from mm/hr to km/hr using the scale you created from measuring the Sun.

The standard unit of CME speed is **km/s**.

7. Convert this velocity from km/hr to km/s.

HINT: there are 3600 seconds in an hour.

Now that we have a value for the speed of the St Patrick's Day 2015 CME, let's calculate what time it should have reached Earth's magnetic field.

8. The Sun is an average of **149,600,000 km** from Earth. Using the speed (km/ hr) you calculated for Image 5, how many hours will it take this CME to reach Earth? Assume a constant speed.

HINT: t = s/v

9. This CME erupted from the Sun at around **02:10 15/03/2015** UT (Wu et al., 2016). Using the number of hours you calculated above, estimate the date and time that this CME should have arrived at Earth. Feel free to use online tools to help you.

Activity 3: Comparing your estimations to the actual St Patrick's Day 2015 event

We will now compare your estimated CME speed and arrival time from **Activity 2** to the actual values for the St Patrick's Day 2015 geomagnetic event.

The actual initial propagation speed of the St Patrick's Day CME was **668 km/s** (Wu et al., 2016).

1. How close was your estimated speed to the actual value?

This CME hit Earth's magnetic field around **04:45 17/03/2015** (Murphy et al., 2018).

2. How close was your estimated CME arrival date/ time to this?

3. Can you think of any assumptions we made or any sources of errors in our calculations that would explain why our estimated values differ from the actual?

The interesting thing about this storm is that it arrived **15 hours ahead** of forecast and was only predicted to be a **G1** level storm (NOAA, 2015)! This shows how important accurate geomagnetic forecasts are.

Additional information about the 3 geomagnetic events you considered in this class can be found in the document **Storm_Event_Information.pdf**.

You can view current geomagnetic conditions and forecasts from the Australian Space Weather Forecasting Centre [here]() and the NOAA [here]().

References

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